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Minimum Mass Optimisation and Laminate Design of a Wing Panel in Element 6 of a Saab 340's Wing

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Abstract

The aim of this study was to design a CFRP wing panel capable of sustaining a specific load that is optimised for minimum weight. Analytical techniques were used to evaluate the wing loading on a Saab 340 to outline the load scenario used for the design of a wing panel. A minimum mass optimisation scheme was derived through analitical methods to obtain geometric dimensions for element 6's wing panel: stiffener width, $b_{st} = 42\ mm$, skin thickness, $t_{sk} = 2.75\ mm$, and stiffener thickness, $t_{st} = 1.625\ mm$. The skin and stiffener laminate stacking sequences were designed using standard industry practices and were determined to be:

t_{sk} : [45/-45/0/45/0/90/0/-45/0_2/45/-45/0_2/-45/0/90/0/45/0/-45/45]

t_{st} : [$\pm 45_1/0_2/\pm 45_1/0_2/90_{1/2}$]$_s$

The final wing panel design achieves a 25.3% weight reduction over the original aluminium panel, with a mass of 6.82 kg/m^2, and maintains structural integrity under the design load with a safety factor of 1.5. Further improvements in analysis and optimisation were discussed, however given the preliminary design application of this analysis, the results were deemed adequate.

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Nomenclature

Symbol	Meaning
CFRP	Carbon Fibre Reinforced Composites
AC	Aerodynamic Centre
COG	Centre of Gravity
$\bar{c}$	Chord length
Subscript *sk*	Skin
Subscript *st*	Stiffener
Subscript *fl*	Flange
Subscript *wb*	Web
Subscript *rf*	Remote Flange
b_{xx}	Width
t_{xx}	Thickness
z_{xx}	Distance from the datum
A_{xx}	Area
l_{box}	Wing box length
t_{box}	Wing box thickness
$b_{sk,op}$	Skin width in open mode
$b_{sk,cl}$	Skin width in closed mode
$b_{sk,max}$	The larger between $b_{sk,op}$ and $b_{sk,cl}$ for the optimisation criteria
M	Bending moment
sf	Safety factor
N_x	Applied load
E_{xx}	Laminate effective modulus
ψ_k	Direction of fibers in the k-th ply
E_{11}	CFRP Elastic Modulus
t_k	Ply thickness
T	Total laminate thickness
K_{xx}	Buckling coefficient
$\overline{Et}$	Panel axial stiffness
$N_{sk,op}$	Buckling load for the open skin
$N_{sk,cl}$	Buckling load for the closed skin
N_{st}	Stiffener buckling load
ε_u	Ultimate strain
$N_{x,mat}$	Material failure load
$\bar{z}$	Neutral axis location
I_{cg}	Moment of area about the centre of gravity
$(EI)_{TOT,NA}$	Total bending stiffness about the neutral axis
$I_{wb,cg}$	Web moment of area about the centre of gravity
l_{rib}	Rib spacing
$N_{x,Euler}$	Euler buckling load
m	Mass
$\bar{t}$	Axial thickness
ρ	Density
RF_{xx}	$\frac{N_{xx}}{N_x}$

Table 1: Symbols used in the report and their meaning

Introduction

This study presents an analytical investigation of a CFRP wing panel used on a Saab 340 with the objective of identifying a configuration that satisfies all structural requirements at minimum mass.

The design brief is provided in Appendix A, and the specific objectives of this work are to:

- Model the wing as a cantilevered beam subjected to discrete loads along its span
- Determine the associated shear forces, bending moments, and torsional loads at key structural locations
- Perform a minimum-mass optimisation of a selected wing panel using analytical methods
- Develop a laminate stacking sequence in accordance with established aerospace design guidelines

Theory

Geometry Definitions

Wing Platform Geometry and Wing Element Definition

To perform the analysis, the Saab 340's wing was divided into seven elements:

Wing element	1 (root)	2 (engine & LG)	3	4	5	6	7 (wingtip)
Length (m)	1.3	1.0	1.3	1.3	1.3	1.8	1.8
Rib Spacing (m)	0.65	-	0.65	0.65	0.65	0.60	0.60
Aluminium mass/area (kg/m^2)	17.5	-	15.0	13.5	11.1	9.1	6.4

Table 2: Element size

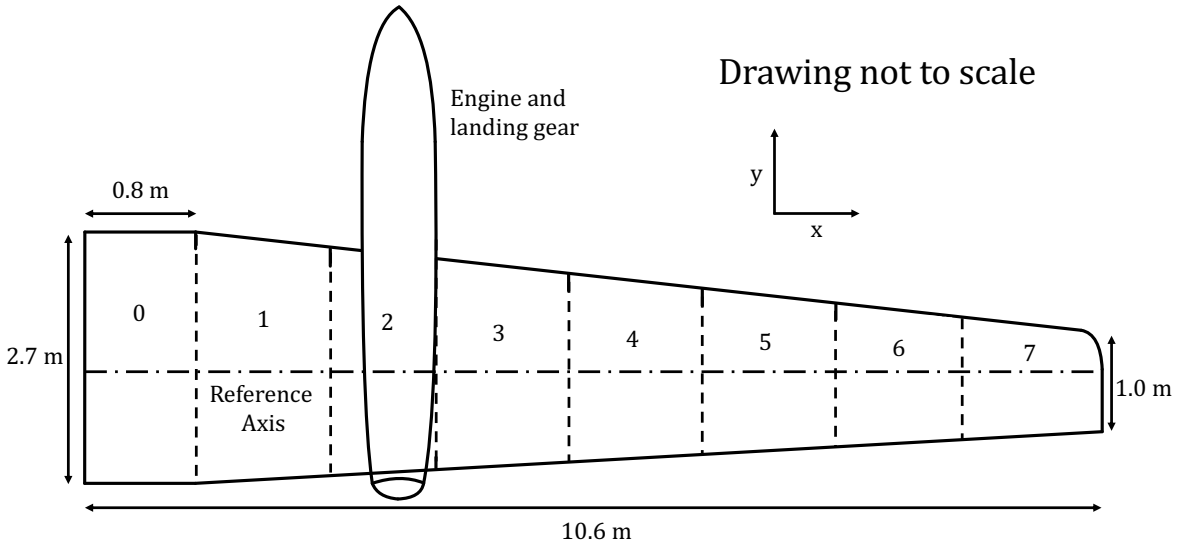


Figure 1: Wing geometry

The MTOW was used to calculate the shear force, bending moment and twist for a scenario at the max-g manoeuvre: 2.71g normal acceleration. The wing weight, comprised of the structural and fuel weights (15% MTOW), acts at 40% chord length from the leading edge, the reference axis (assumed to be unswept) and wing box centre are also located at 40% chord length, whilst lift acts at 25% chord[2]. The wing element chosen for the optimisation was element 6.

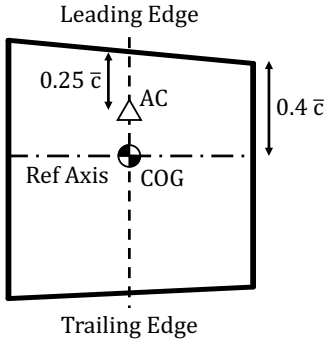


Figure 2: Wing element definition (element 6 shown)

Hat-Stiffened Compression Panel Geometry

Analysis was performed on a single stiffener panel out of a repeating pattern, the panel's geometry is shown below.

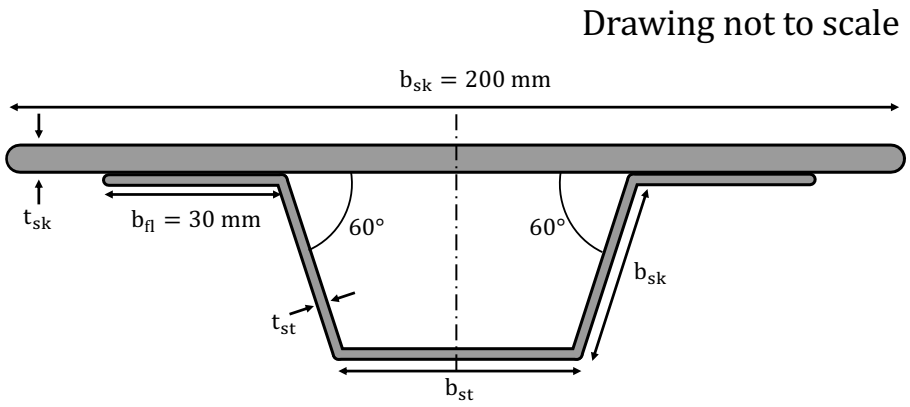


Figure 3: Hat-Stiffened panel geometry

Skin and Stiffener Buckling Loads

It was assumed that the upper and lower panels carry the full bending moment but carry no shear and are uniformly strained. The applied panel load is a function of the bending moment for a given element[1]:

$$N_x = \frac{M \times sf}{l_{box} \times t_{box}} \,. \tag{1}$$

Both the skin and stiffener were made from CFRP and had a desired ply percentage breakdown of [44/44/12] and [60/30/10] respectively[2]. The effective modulus of the panels was calculated using Krenchel's approximation for an n-play CFRP laminate[1],

$$E_{xx} = \sum_{k=1}^{N} (\cos \psi_k)^4 E_{11} \frac{t_k}{T} \, . \tag{2}$$

The axial (or smeared) stiffness of the panel-stiffener combination is given by[1]

$$\overline{Et} = E_{sk} t_{sk} + E_{st} \frac{A_{st}}{b_{sk}} \, . \tag{3}$$

Thus, the first four buckling loads can be computed[1]:

$$N_{sk,op} = \overline{Et} \times K_{sk} \times \left(\frac{t_{sk}}{b_{sk,op}} \right)^2 , \tag{4}$$

$$N_{sk,cl} = \overline{Et} \times K_{sk} \times \left(\frac{t_{sk}}{b_{sk,cl}} \right)^2 , \tag{5}$$

$$N_{st} = \overline{Et} \times K_{st} \times \left(\frac{t_{st}}{b_{st}} \right)^2 , \tag{6}$$

$$N_{x,mat} = \overline{Et} \times \varepsilon_u \, . \tag{7}$$

To calculate the Euler buckling load, the geometry's neutral axis must first be found. The calculation was performed with the skin mid-height as the datum[1].

$$\bar{z} = \frac{\sum EAz}{\sum EA} \tag{8}$$

$$\bar{z} = \frac{2\left(E_{st} A_{fl} z_{fl}\right) + 2(E_{st} A_{wb} z_{wb}) + E_{st} A_{rf} z_{rf}}{2\left(E_{st} A_{fl}\right) + 2(E_{st} A_{wb}) + E_{st} A_{rf} + E_{sk} A_{sk}} \tag{9}$$

The overall buckling load is found with[1]

$$(EI)_{TOT,NA} = \sum EAy^2 + EI_{cg} \, , \tag{10}$$

$$(EI)_{TOT,NA} = E_{sk}A_{sk}\bar{z}^2 + E_{sk}\frac{b_{sk}{t_{sk}}^3}{12} + 2\left(E_{st}A_{fl}\left(\bar{z} - z_{fl}\right)^2 + E_{st}\frac{b_{fl}{t_{st}}^3}{12}\right) +$$
$$E_{st}A_{rf}\left(z_{rf} - \bar{z}\right)^2 + E_{st}\frac{b_{st}{t_{st}}^3}{12} + 2\left(E_{st}A_{wb}(z_{wb} - \bar{z})^2 + E_{st}I_{wb,cg}\right). \quad (11)$$

Therefore the Euler buckling load is[1]

$$N_{x,Euler} = \frac{\pi^2 (EI)_{TOT,NA}}{{l_{rib}}^2 \times b_{sk}}\ . \quad (12)$$

Optimisation Scheme

Optimality Criteria

The first optimality criteria was defined by equating the skin and stiffener buckling loads, ensuring both components reach their critical loads simultaneously[2],

$$K_{sk}\left(\frac{t_{sk}}{b_{sk,max}}\right)^2 = K_{st}\left(\frac{t_{sk}}{b_{st}}\right)^2. \quad (13)$$

The second was defined by equating the applied load and the stiffener buckling load[2],

$$N_x = K_{st}\left(\frac{t_{st}}{b_{st}}\right)^2. \quad (14)$$

Using **(13)**, **(14)**, and **(3)**, t_{st} can be expressed in terms of b_{st}, and t_{sk} can be expressed in terms of b_{st} and t_{st}

$$t_{st} = \sqrt[3]{\frac{N_x {b_{st}}^2}{K_{st}\left(\frac{E_{sk}b_{sk,max}\sqrt{\frac{K_{st}}{K_{sk}}}}{b_{st}} + \frac{E_{st}\left(3b_{st} + 2b_{fl}\right)}{b_{sk}}\right)}}\ , \quad (15)$$

$$t_{sk} = b_{sk,max}\frac{t_{st}}{b_{st}}\sqrt{\frac{K_{st}}{K_{sk}}}\ . \quad (16)$$

(15), and **(16)**, can be plotted to show the values of t_{st} and t_{sk} over a given b_{st}.

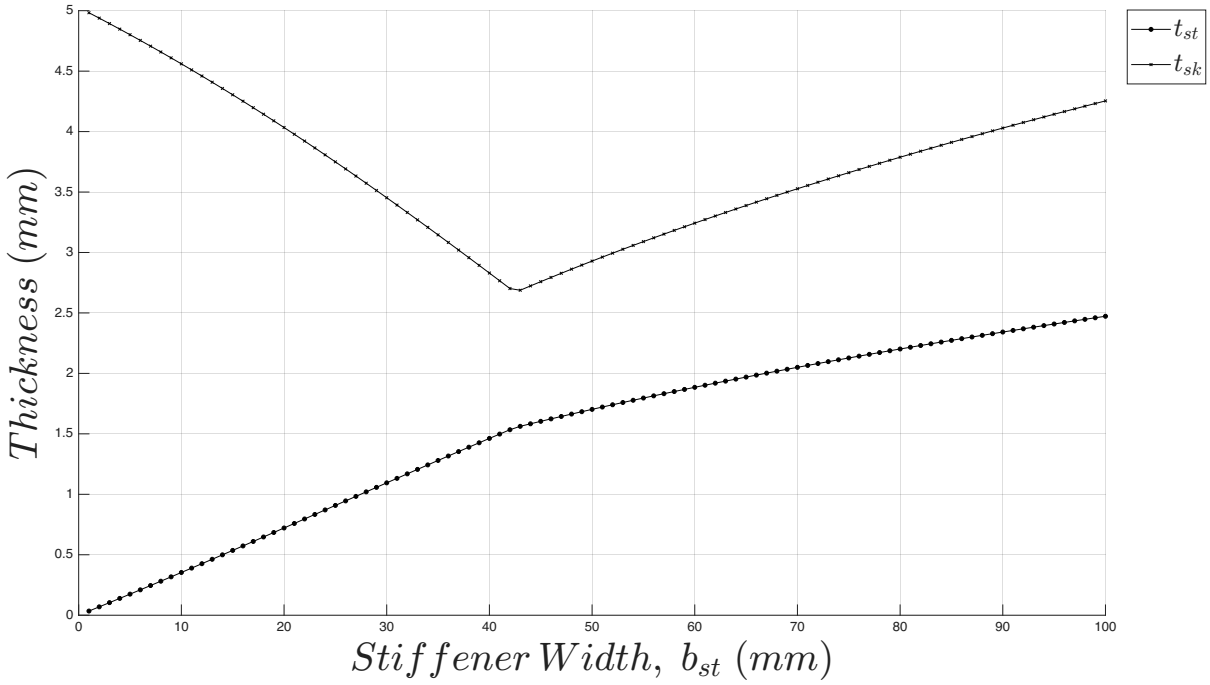


Figure 4: Skin (symbol) and Stiffener (symbol) thickness vs Stiffener width

Mass Calculation

The mass of the element can be calculated with[2],

$$m = \bar{t} \times \rho \,, \tag{17}$$

where the smeared thickness, $\bar{t}$, is given by[2]

$$\bar{t} = t_{sk} + \frac{A_{st}}{b_{sk}} \,. \tag{18}$$

These equations relate mass to b_{st}, thus **(17)** can be plotted to show mass over a given b_{st}.

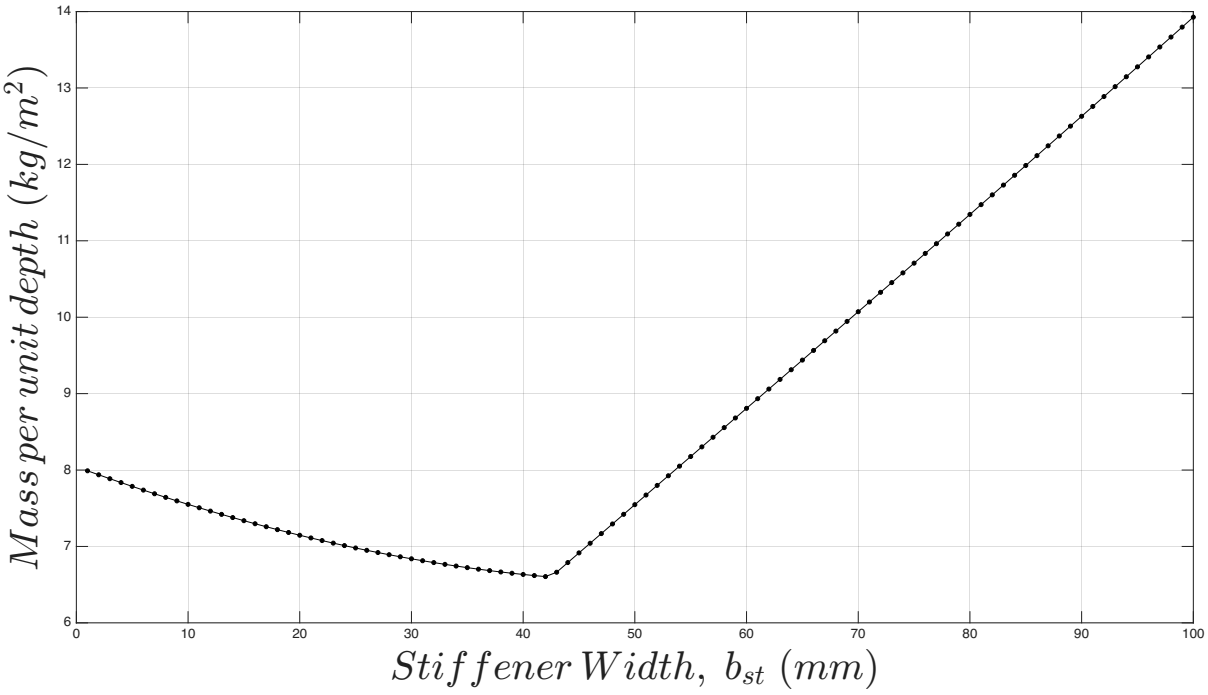


Figure 5: Element mass vs Stiffener width

Reserve Factor Conditions

Reserve factors are defined as the buckling loads devided by the applied load. This normalises the buckling modes to the applied load. For a satisfactory panel design, the following reserve factor conditions had to be met[2]:

$$RF_{sk,op} \geq 1, \qquad RF_{sk,cl} \geq 1, \qquad RF_{st} \geq 1, \qquad RF_{x,mat} \geq 1, \qquad RF_{x,Euler} \geq 1.5 \,. \tag{19}$$

Mass Optimisation

MATLAB was used to find the lightest panel design that satisfied all the reserve factor conditions. The data was loaded and an array of b_{st} values, ranging from 1 to 100 mm with steps of 1 mm, was created. All calculations were performed with the given value of b_{st} as all the variables used to calculate the reserve factors and panel mass either had a fixed value or were dependant on b_{st}. The outputs were reserve factors and mass arrays. A range of b_{st} values that satisfied all the reserve factor conditions was then identified and the lowest mass in that range was found. This optimisation approach was chosen over an iterative approach because it explored all the potential configurations with a given b_{st} value, allowing the best one to be selected instead of the first that satisfied the conditions. The flowchart below shows the optimisations process.

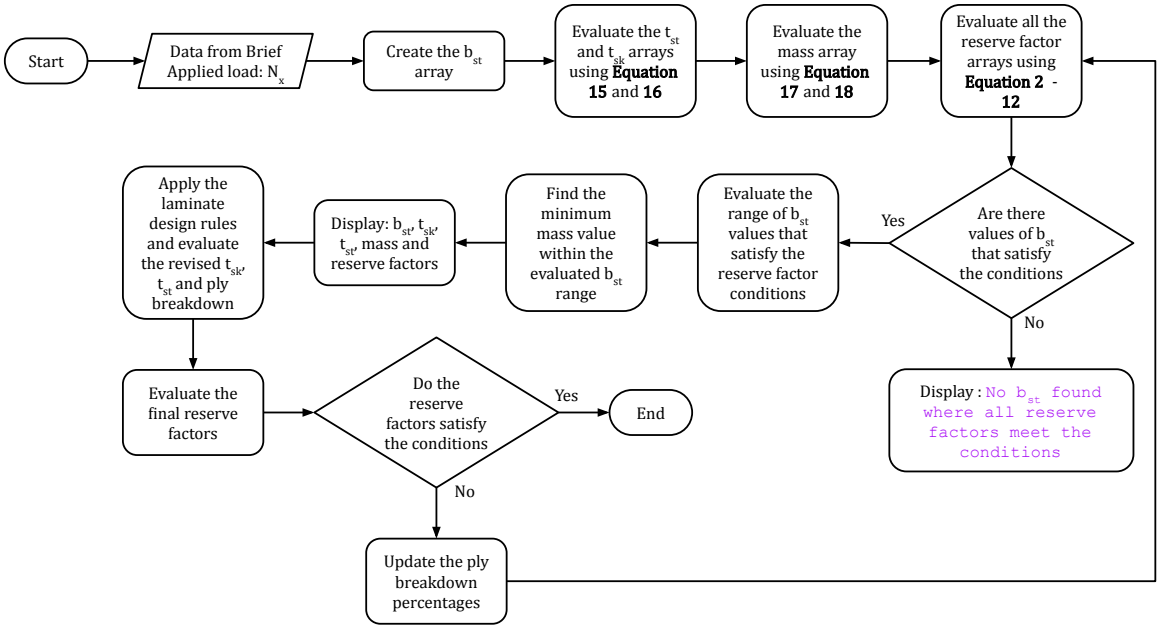


Figure 6: Optimisation Code

Note: Flowchart shows consequences of laminate design that will be discussed in the following section

Laminate Design

Once the final t_{st} and t_{sk} values were found, laminate design was considered. The optimised t_{st} and t_{sk} values were adjusted to be multiples of the ply thickness, which was given as 0.125mm[2]. The number of plies in each orientation was then calculated using the desired ply percentage breakdown. The industry standard composite stacking sequence rules, found in Appendix B, were then implemented to find an appropriate laminate design.

The reserve factors were then recalculated with the revised thicknesses and ply percentage breakdowns to ensure the final design still met all the conditions. If the conditions were not met, the optimisation process would be repeated with the new ply percentage breakdown evaluated after laminate design.

Results

Aircraft Shear, Bending Moment and Twist Graphs

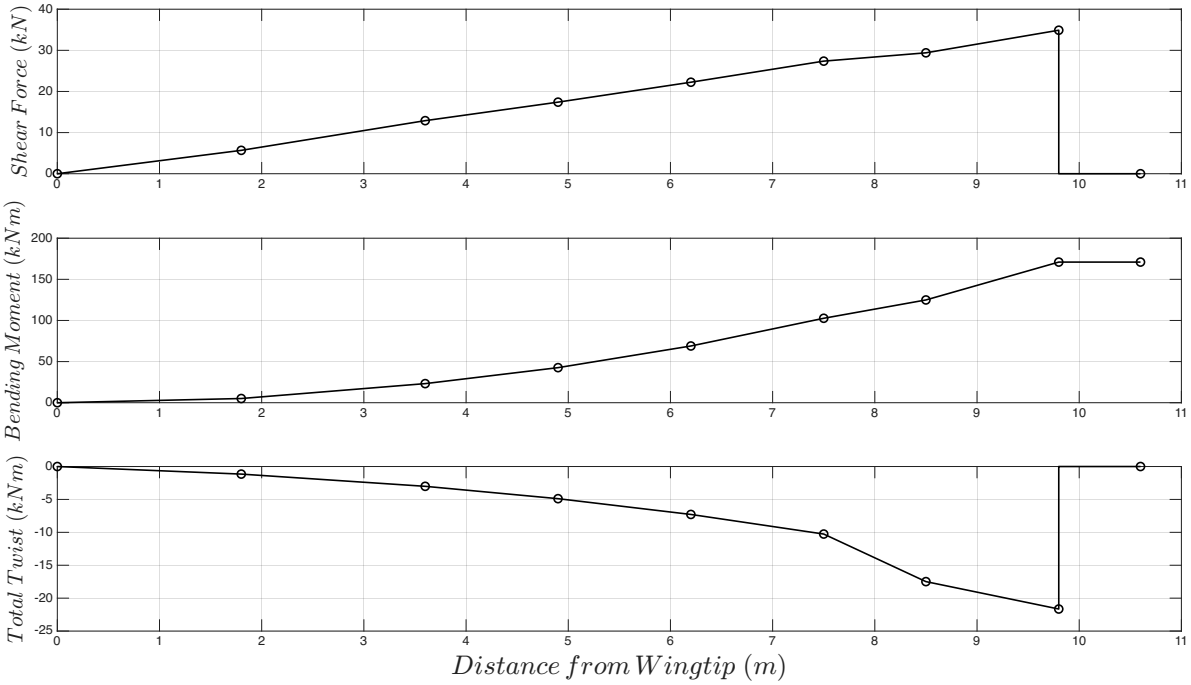


Figure 7: Aircraft Shear Force, Bending Moment and Twist Diagrams for the 1g case

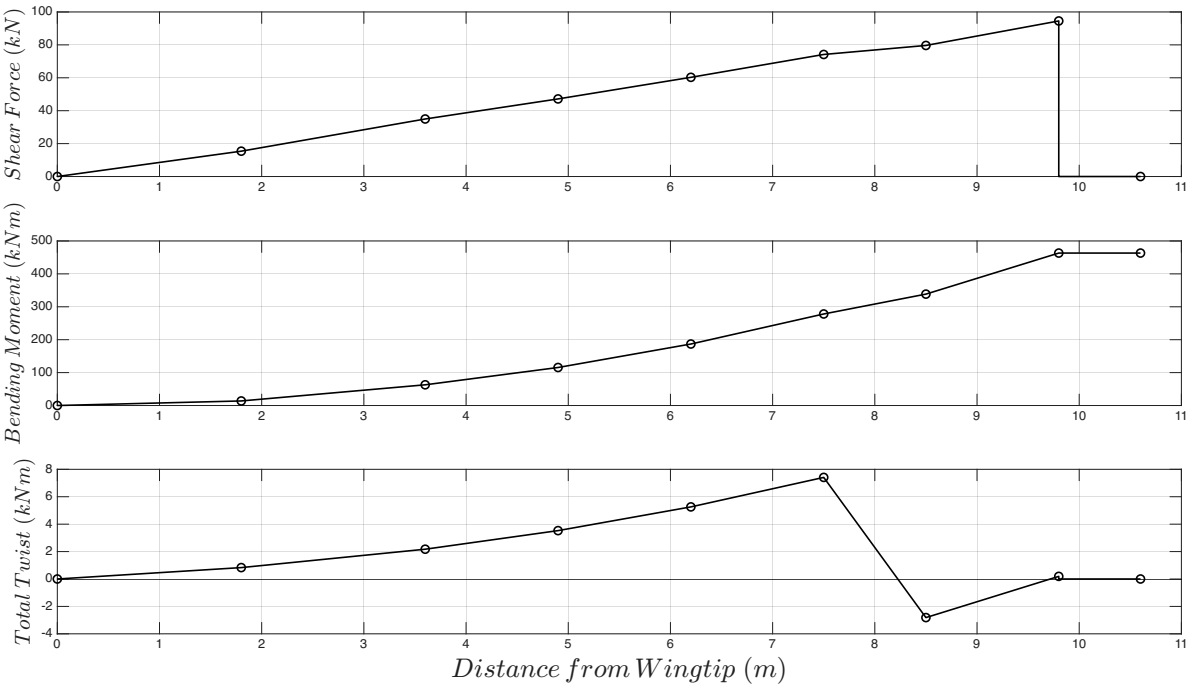


Figure 8: Aircraft Shear Force, Bending Moment and Twist Diagrams for the 1g case

Full calculation breakdowns can be found in Appendix C and D.

Wing Panel Element 6 Optimisation

N_x was calculated to be 0.7008 kN/mm using **(1)** and the bending moment value from element 6.

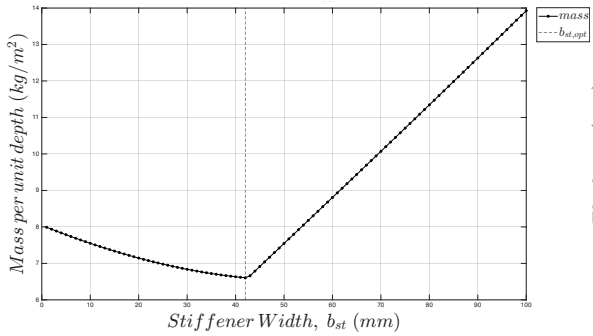


Figure 9: Cross-Section Mass vs Stiffener Width

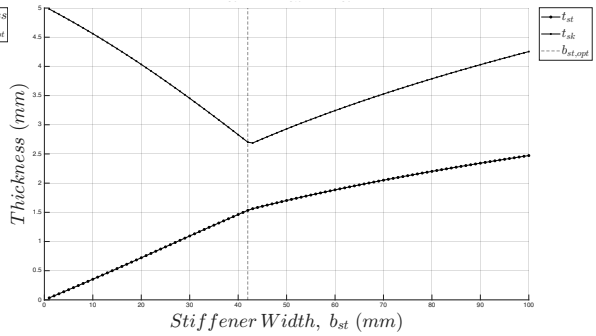


Figure 10: Stiffener & Skin Thickness vs Stiffener Width

Optimised b_{st} value shown by a vertical dashed line.

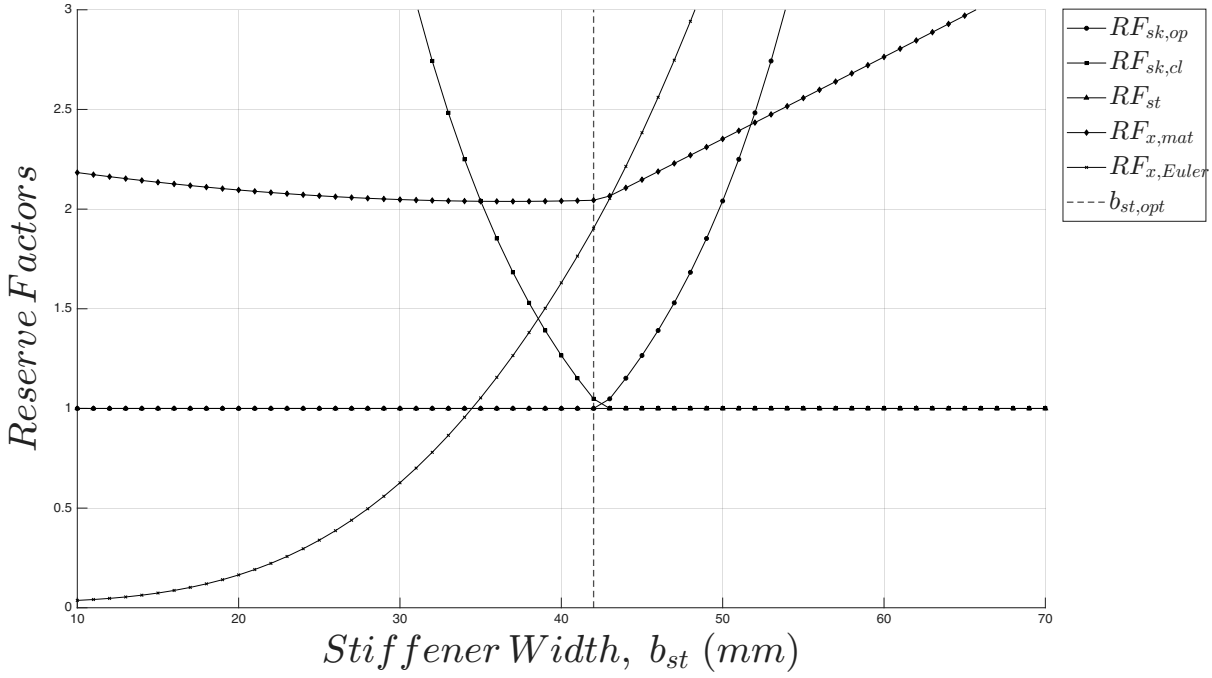


Figure 11: Compression Panel Reserve Factors vs Stiffener Width

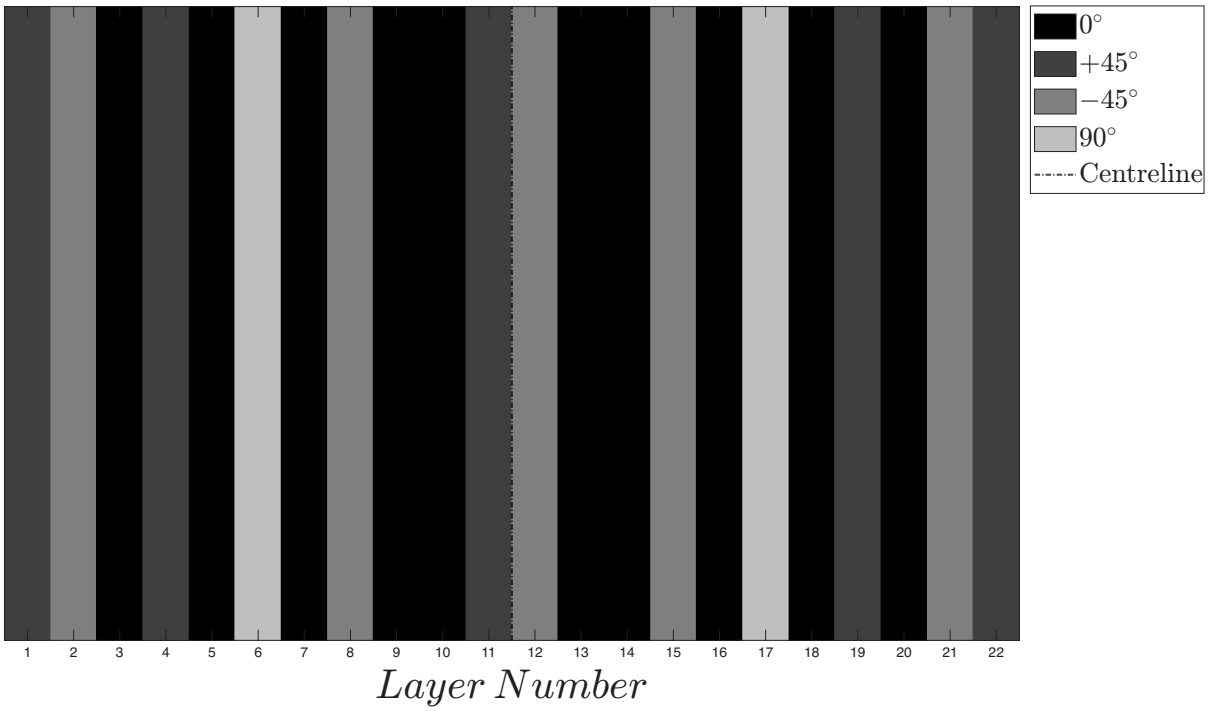


Figure 12: Skin Laminate Sequence

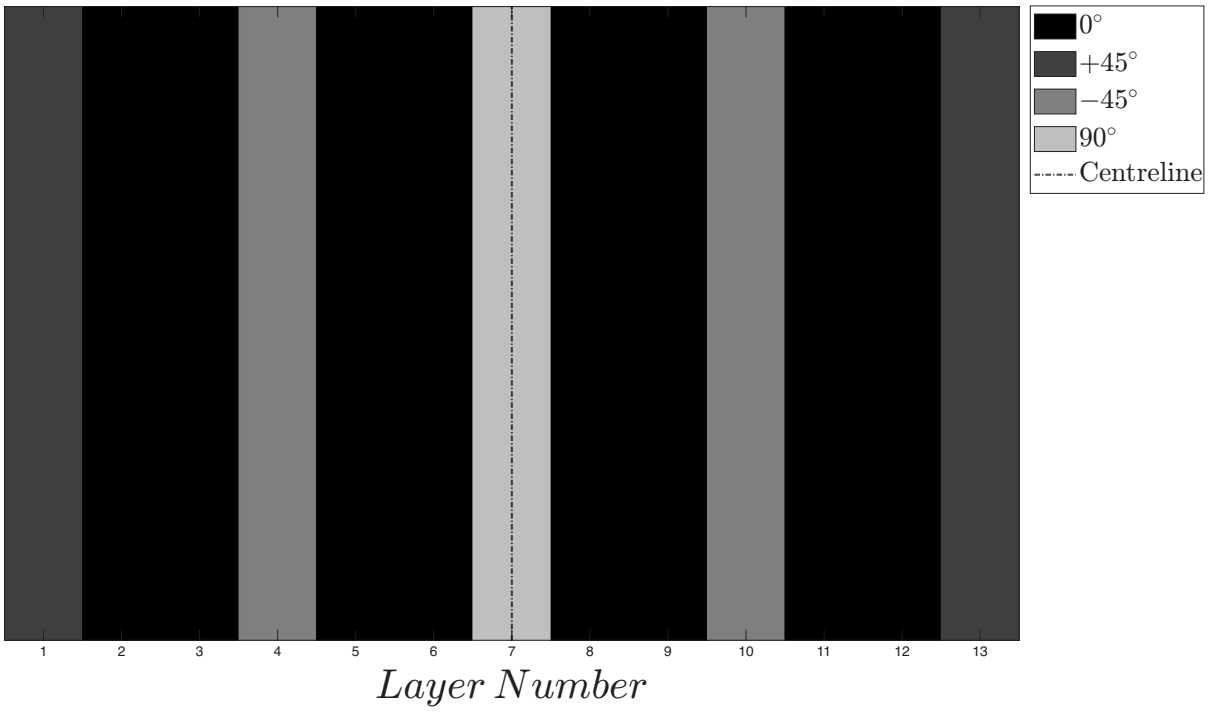


Figure 13: Stiffener Laminate Sequence

Thickness	Laminate Stacking Sequence
t_{sk}	[45/-45/0/45/0/90/0/-45/0_2/45/-45/0_2/-45/0/90/0/45/0/-45/45]*
t_{st}	$[\pm 45_1/0_2/\pm 45_1/0_2/90_{1/2}]_s$

Table 4: Laminate Stacking Sequences

*t_{sk} stacking sequence is not symmetric, discussed in the Discussion section

Design	b_{st}	t_{sk}	t_{st}	$RF_{sk,op}$	$RF_{sk,cl}$	RF_{st}	$RF_{x,mat}$	$RF_{x,Euler}$	t_{sk} Ply breakdown	t_{st} Ply breakdown
Optimum	42.0	2.70	1.50	1.00	1.04	1.00	2.04	1.90	44/44/12	60/30/10
Laminate Design	42.0	2.75	1.625	1.10	1.16	1.20	2.18	2.06	45/45/10	62/30/08

Table 4: Comparison of Design Configurations (lengths in mm)

Material	Configuration	Weight per area (kg/m²)	% Decrease
Aluminium	Given	9.1	Datum
CFRP	Optimum	6.6	27.5
CFRP	Laminate Design	6.8	25.3

Table 5: Comparison of Weights and % decrease

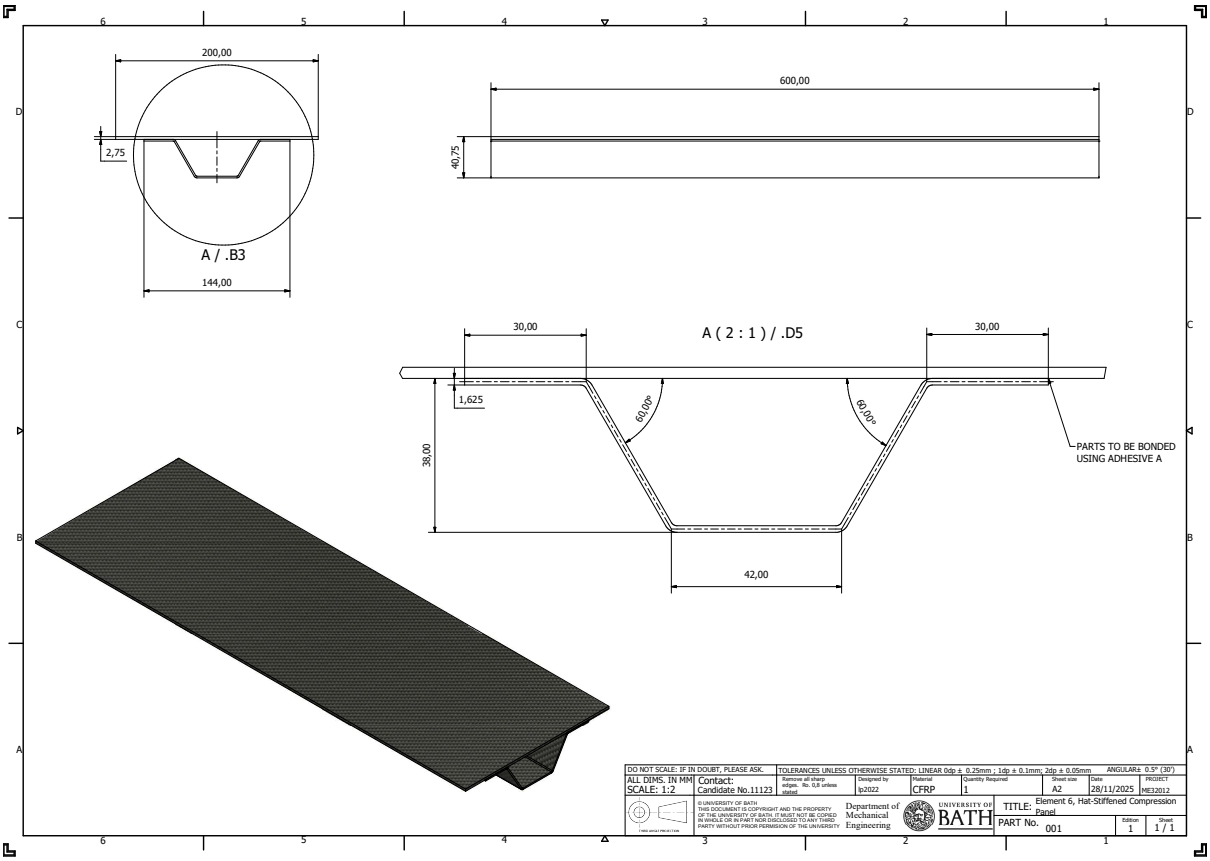


Figure 14: Compression Panel Engineering Drawing

Discussion

Wing Loading

Shear force and bending moment increased linearly and quadratically respectively, following cantilever-beam behaviour. Shear returned to zero at the wing root (9.8 m) due to the opposing fuselage reaction force, causing the bending moment to remain constant within element 0. Twist about the neutral axis is caused by the moment from lift at the aerodynamic centre, the pitching moment, and off-axis masses. In the 1g case, the negative pitching moment (and engines) produced negative twist, while in the 2.71g case, lift partly counteracted this but the increased engine weight still caused negative local twist. Twist returned to zero at the root due to fuselage restraint.

Because loads increase toward the root, the structure must strengthen accordingly, therefore tailoring it to local spanwise is necessary to minimise weight, justifing the optimisation process.

Hat-Stiffened Compression Panel

The success of the optimisation scheme was validated through the inspection of the results which met the reserve factor criteria outlined in the theory section. Furthermore, the final panel geometry was the lightest possible configuration, as its weight sat at the minimum point of the mass vs b_{st} curve (**Figure 9**), decreasing by 25.3% of the original aluminium panel's weight. A b_{st} step size of 1mm was deemed appropiate as the changes in thicknesses for each b_{st} step was smaller than 0.125mm.

The behaviour of RF_{st}, $RF_{sk,op}$ and $RF_{sk,cl}$ reflect the conditions set by **(13)** as the skin buckling load reserve factor with the larger between $b_{sk,cl}$ and $b_{sk,op}$ is equal to RF_{st} for that b_{st} value. Furthemore, $RF_{st} = 1$ for all b_{st} values, reflecting the condition set by **(14)**. The Euler buckling mode shows the most dependacy to b_{st}, for low values, Euler buckling is the only critical mode, however, it increases exponentially as b_{st} becomes larger. This suggests Euler buckling must be examined more carefully when designing a support structure. A weaker, therefore potentially lighter and cheaper, material could have been considered for this panel as $RF_{x,mat}$ has a high value for every b_{st}. This however might lead to greater skin and stiffener thicknesses being needed, therefore negating any possible cost or weight advantages.

The thicknesses and number of layers restricted the options availible for different stacking configurations and differed the final panel's ply percentage breakdown from the desired breakdown. The laminate designs respect the geometric guidelines stated by the brief but do not comply with every guideline outlined in Appendix B. The skin laminate is not symmetrical as there are an odd number of +45° and -45° plies. This was the result of following the desired ply percentage distribution with the optimised skin thickness. For a design to be symmetrical, the number of $\pm 45°$ layers must be divisible by four. Unsymmetrical laminates cause stretching and bending to become coupled, so loads produce unintended deformation and reduce structural efficiency[3]. To produce a symmetric laminate, six $+45°$ and four -45° plies could have been used, but an unbalanced laminate is worse than an unsymmetrical laminate. Alternatively, two $\pm 45°$ plies could

have been added, but that would have increased mass by 0.4kg and deviated from the desired ply distribtion. A potential solution would have been to increase the number of $\pm 45°$ layers to twelve and decrease the 0° layers to eight. That would deviate the ply distribution from the desired distribution but another set of calculations proves the design would still be adequate.

Design	b_{st}	t_{sk}	t_{st}	$RF_{sk,op}$	$RF_{sk,cl}$	RF_{st}	$RF_{x,mat}$	$RF_{x,Euler}$	t_{sk} Ply breakdown	t_{st} Ply breakdown
Laminate Design	42.0	2.75	1.625	1.10	1.16	1.20	2.18	2.06	45/45/10	62/30/08
Revised	42.0	2.75	1.625	1.02	1.07	1.11	2.02	2.00	36/55/09	62/30/08

Table 6: Comparison of Design Configurations (lengths in mm)

Doing this however would change the strength of the laminate in a given direction, the ideal ply distribution was chosen for the panel's specific load case so further analysis would be required to check whether this new distribution is still acceptable. The +45° and -45° plies were therefore placed in the centreline as that limits the ill effects of asymmetrical laminates and the optimised design was used[1].

The stiffener laminate design did not respect the sixth rule, having 8% of the distribution being at 90°, leading to a structurally biased material. This is a consequence of the few layer numbers due to the small thickness. The solutions to this problem would have been similar to those outlined earlier: increase in thickness or a change in laminate distribution, however, given the ideal ply distribution already showed bias in strength for a given direction, the design can be left unchanged.

Limitations, Assumptions and Further Analysis

Analytical methods were used in this analysis, requiring several simplifications to the geometry and applied loads. Forces were assumed to vary linearly between element boundaries, which provides a reasonable approximation but reduces accuracy, particularly in regions of high load variation such as element 2, where the engine and landing gear are located. Increasing the number of elements would improve accuracy but would also make the analytical approach significantly more time-consuming. For greater fidelity, numerical methods such as FEA would be required.

The aircraft geometry was also simplified. The neutral axis was assumed to be unswept, despite the actual wing sweep shown in **Figure 1**. Wing spars and ribs were not included, and the assumption that the skin carries no shear force is not valid. Additionally, the upper and lower panels were assumed to experience uniform strain, which may not be true across the entire wing. Different material properties and possible stress concentrations due to geometry were neglected. Manufacturing defects were also not considered, although the applied safety factor of 1.5 helps partially account for these uncertainties.

The optimisation method was effective in simplifying the design process but was derived from idealised theory. It also limited the range of design configurations explored, and there may be lighter solutions that still satisfy the reserve factor requirements. A full

three-parameter search, with b_{st} incremented in 1 mm steps and t_{st} and t_{sk} in 0.125 mm steps to reflect ply thickness constraints, could identify these alternatives. However, this approach would be more computationally demanding and difficult to justify at the preliminary design stage.

Finally, the effective moduli E_{sk} and E_{st} were held constant, even though changes in thickness would alter the ply distribution and therefore the stiffness. Incorporating these effects would improve accuracy but add significant complexity for a relatively small gain in precision, which is not justified for the given time constraints and an initial design study.

Conclusions

The results from the shear force, bending moment and twist calculations respected theory and provided a base for the wing panel optimisation process. The implemented optimisation resulted in an adequate light-weight panel for the given loading conditions. The panel specifications are shown below.

$\boldsymbol{b_{st}}$	42.0 mm	$\boldsymbol{RF_{sk,op}}$	1.10
$\boldsymbol{t_{sk}}$	2.75 mm	$\boldsymbol{RF_{sk,cl}}$	1.16
$\boldsymbol{t_{st}}$	1.625 mm	$\boldsymbol{RF_{st}}$	1.2
$\boldsymbol{t_{sk}}$ **Ply breakdown**	45/45/10	$\boldsymbol{RF_{x,mat}}$	2.18
$\boldsymbol{t_{st}}$ **Ply breakdown**	62/30/08	$\boldsymbol{RF_{x,Euler}}$	2.06
Weight	6.8 kg/m^2	**% Optimisation from original panel**	25.3%

Table 7: Final Panel Specifications

Some deviation from standard industry guidelines, mainly rules two and six, resulted during laminate design due to the optimised geometry of the panel. These deviations however were minor, and their ill effects are limited. Therefore a change in geometry from the optimised design was not deemed necessary. Assumptions were made to accommodate an analytical approach; therefore the accuracy of this model is limited. Future work should rely on more complex analysis such as FEA and physical testing with the full wing geometry being considered, however given the scope of this study is preliminary analysis and design, the methods used were adequate.

References

1. Butler, R. (2025). *Aircraft Loads and Wing Panel Design*. Lecture Notes
2. Butler, R. (2025). *Wing Design Coursework*. Brief
3. Jones, R.M. (1999). *Mechanics of Composite Materials*. 2nd Edition ed. Chem/Mats-Sci/E.

Appendix A: Wing Panel Design Brief

University of Bath **ME32012 Aircraft Performance and Design**
Department of Mechanical Engineering **Coursework**

FLIGHT TEST REPORT II
SAAB 340 Wing Panel Design

You are asked to evaluate the benefit of replacing an aluminium panel in the upper surface of the SAAB 340 wing with a composite panel.

Part (i)
Plot the graphs for limit shear force, bending moment and torque along the wing semi-span (in reference axes) considering a maximum-*g* manoeuvre case at V_C. Assume that the aircraft (GA pages 3-5, approximate dimensions on pages 1-2 to be used) is at MTOW with 25.8 kN fuel and that the lift produced per unit wing area is constant, and equal to *W/S*, where *W* is aircraft weight and *S* is total wing area. Lift acts at quarter chord.

Part (ii)
Assuming local buckling coefficients for skin and stiffeners of 2.23 and 1.65, respectively, design the idealised carbon fibre reinforced plastic (CFRP composite) hat-stiffened compression panel *AC* of Fig. 1, with detailed cross-section shown in Fig. 2, at the inboard side of one of the spanwise elements specified in Table 1, according to your allocated group. Your design should include two 30 mm-wide, balanced-symmetric flanges for attachment of each stiffener to the skin.

You should also assume: (1) that the upper and lower panels carry the full bending moment but carry no shear; (2) that the panels are uniformly strained; (3) stiffener spacing (b_{sk}) of 200 mm respectively, and (4) that spars *AB* and *CD* are located at 20 and 60% chord, respectively. Rib spacings for each wing element are given in Table 1.

Your report should give evidence of optimisation to achieve minimum mass, given that the following should not occur before *ultimate* load: (1) local skin buckling between stiffeners[1]; (2) local skin buckling underneath stiffeners[2]; (3) local stiffener buckling[3]; (4) overall buckling[4], and (5) material failure. You should take stacking sequence design rules into account.

Wing Element	**1 < (root)**	**2 (engine & LG)**	**3**	**4**	**5**	**6**	**7 (tip) >**
Length (m)	1.3	1.0	1.3	1.3	1.3	1.8	1.8
Group	A	-	B	C	D	E	F
Rib spacing (m)	0.65	-	0.65	0.65	0.65	0.60	0.60
Aluminium mass/area (kg/m^2)	17.5	-	15.0	13.5	11.1	9.1	6.4

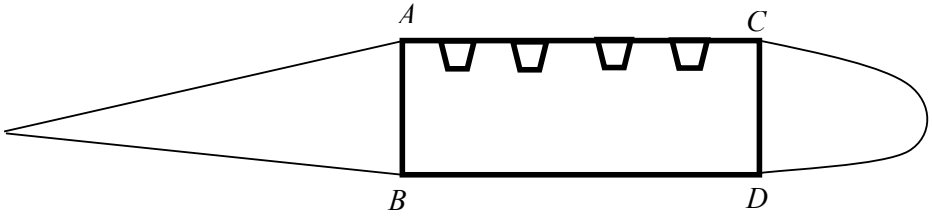
Table 1. Element length, groups for Part (ii) and rib spacing.

Notes: 1. Assume effective width (b_{eff}) is distance between flange centres; 2. Assume $b_{eff} = 2\ b_{st}$; 3. Assume $b_{eff} = b_{st}$.; 4. To allow for the reduced overall buckling load of a simply-supported stiffened panel, you should ensure a Reserve Factor greater than 1.50 for the panel Euler buckling load.

SAAB 340 Data

- Max. Take Off Weight (MTOW) = 130.37 kN
- Total Wing Area S = 40.58 m^2 (including rectangular wing area within fuselage)
- Body side is at 0.8 m from aircraft centre line and chord is constant through fuselage
- Body side chord = 2.7 m; tip chord = 1.0 m
- Wing reference axis (40 % chord) is unswept
- Use 7 spanwise wing elements of length shown in Table 1
- Wing thickness/chord ratio = 12 %
- Wing weight (including systems) = 15 % MTOW, and acts in proportion to wing area at 40 % chord
- Element 2 comprises the engine and main landing gear, weighing an additional 4.5 kN with c.g. 1.0 m in front of the centre of the wing box. [Assume no thrust.]
- Fuel weight calculated from fuel vol., with one tank per element; elements 2, 6 and 7 are empty and tanks positioned between the front spar and rear spar. Fuel is emptied from inboard tanks first.
- Fuel specific gravity = 0.9 (0.9 kg/litre)
- V_c = 140 m/s (EAS) and C_{MO} = -0.08
- Elastic modulus of CFRP plies in fibre direction = 130 kN/mm^2 (Assume Krenchel's approximation for laminate stiffness and ply percentage breakdowns of 44/44/12 and 60/30/10 for skin and stiffener laminates, respectively. Ply thickness = 0.125 mm)
- Maximum strain of skin and stiffeners is 0.0045 m/m.
- Density of CFRP = 1600 kg/m^3

Fig. 1 Schematic cross-section of wing



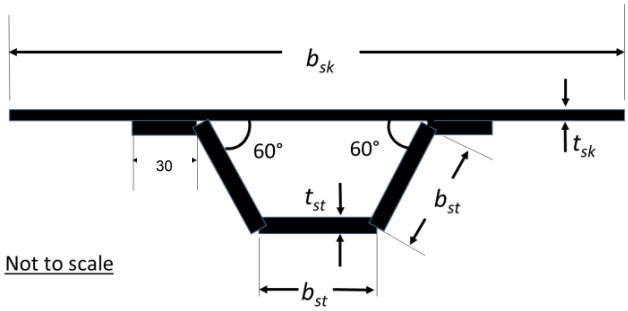


Fig. 2 Repeating portion of hat-stiffened panel

Word limit = 1000 words (excluding figures, equations & tables); concise reports are encouraged.
Hand in – 4.00pm Friday 12th December 2025

Appendix B: Industry Standard Composite Stacking Sequence Rules[1]

7.2 Stacking sequence rules for composite laminates

Layers are orientated with fibres in a given direction defined by the fibre angle ψ_k of each layer k where $\psi_k = 0°$ is parallel to the reference axis. The following rules are applied in order to create laminates with practical stacking sequences.

1. Fibre angles are normally restricted to 0°, 90°, +45° and -45°.
2. For most practical purposes, laminates are both balanced (i.e. they have as many +45° layers as they have -45° layers) and symmetric about their mid-plane.
3. Grouping of more than 3 layers of the same orientation is not normally permitted - to avoid large inter-lamina stresses.
4. A +45°/-45° pair is normally positioned at the surface to increase damage tolerance.
5. +45° and -45° layers are usually placed next to each other in order to minimise elastic bend-twist coupling.
6. Normally, at least 10% of each fibre angle (0°, 90°, +45° and -45°) is required for robust design.

Laminates are tailored according to application. For example, target (0°/±45°/90°) ply percentages for stiffeners, skins and spars are typically 60/30/10, 44/44/12 and 10/80/10, respectively.

Example

Suppose we are required to create a 6.2mm thick laminate consisting of 60% of 0° layers, 30% of ±45° layers and 10% of 90° layers, i.e. a 60/30/10 laminate. The target thickness for each orientation is then 3.72mm/1.86mm/0.62mm. If the layers are 0.25mm thick, rounding up gives 15/8/3 layers, although in order to allow for symmetry we would need to use 16 layers of 0° orientation and 4 layers of 90° orientation. One stacking sequence that satisfies the above rules would be:

$[\pm 45/0_3/\pm 45/0_3/90/0_2/90]_S$

Appendix C: 1g Shear Force, Bending Moment and Twist Calculations

	Wingtip					Engine		Fuselage
Element	7	6	5	4	3	2	1	0
Width (m)	**1.8**	**1.8**	**1.3**	**1.3**	**1.3**	**1.0**	**1.3**	**0.8**
Distance from tip to Fuselage (m)	1.8	3.6	4.9	6.2	7.5	8.5	9.8	10.6
Distance to the Mid-chord (m)	0.9	0.9	0.65	0.65	0.65	0.5	0.65	0.4
Mid-chord distance from Fuselage (m)	0.9	2.7	4.25	5.55	6.85	8	9.15	10.2
End-chord length (m)	1.31	1.62	1.85	2.08	2.30	2.47	2.70	2.70
Mid-chord length (m)	1.16	1.47	1.74	1.96	2.19	2.39	2.59	2.70
Area of the strip (m^2)	2.08	2.64	2.26	2.55	2.84	2.39	3.36	2.16
Thickness (m)	0.14	0.18	0.21	0.24	0.26	0.29	0.31	0.32
Fuel tank length (m)	0.46	0.59	0.69	0.79	0.88	0.96	1.03	1.08
Fuel tank volume (m^3)	0.00	0.00	0.19	0.24	0.30	0.00	0.42	0.28
Fuel tank volume (m^3) - Integration	**0.000**	**0.000**	**0.189**	**0.241**	**0.299**	**0.000**	**0.418**	**0.280**
Fuel volume inside tank (m^3)	0.000	0.000	0.189	0.241	0.299	0.000	0.418	0.280
Fuel mass inside tank (kg)	0.00	0.00	169.73	216.59	269.16	0.00	376.16	251.94
Fuel weight inside tank (kN)	**0.00**	**0.00**	**1.67**	**2.12**	**2.64**	**0.00**	**3.69**	**2.47**
Empty wing weight (kN)	**1.00**	**1.27**	**1.09**	**1.23**	**1.37**	**1.15**	**1.62**	**1.04**
Extra weights (kN)	0.00	0.00	0.00	0.00	0.00	4.50	0.00	0.00
Total wing weight (kN)	1.00	1.27	2.75	3.35	4.01	5.65	5.31	3.51
Lift (kN)	**6.69**	**8.49**	**7.26**	**8.20**	**9.14**	**7.67**	**10.81**	**6.94**
Resultant force (kN)	**5.68**	**7.22**	**4.50**	**4.84**	**5.13**	**2.02**	**5.49**	**3.43**
Shear force (kN)	**5.68**	**12.90**	**17.40**	**22.25**	**27.37**	**29.39**	**34.89**	**0.00**
Bending Moment (kNm)	**5.11**	**21.84**	**41.54**	**67.31**	**99.56**	**127.94**	**169.73**	**0.00**
Reference axis location	0.46	0.59	0.69	0.79	0.88	0.96	1.03	1.08
Aerodynamic centre location	0.29	0.37	0.43	0.49	0.55	0.60	0.65	0.68
Pitching moment about ref axis (Nm)	-1151.23	-1857.05	-1877.36	-2396.39	-2978.69	-7228.11	-4163.90	0.00
Local Twist (kNm)	-1.15	-1.86	-1.88	-2.40	-2.98	-7.23	-4.16	0.00
Total Twist (kNm)	**-1.15**	**-3.01**	**-4.89**	**-7.28**	**-10.26**	**-17.49**	**-21.65**	**0.00**

Table 8: Analysis of Shear Force, Bending Moment and Twist on the wing for the 1g case

Appendix D: 2.71g Shear Force, Bending Moment and Twist Calculations

	Wingtip					Engine		Fuselage
Element	7	6	5	4	3	2	1	0
Width (m)	1.8	1.8	1.3	1.3	1.3	1.0	1.3	0.8
Distance from tip to Fuselage (m)	1.8	3.6	4.9	6.2	7.5	8.5	9.8	10.6
Distance to the Mid-chord (m)	0.9	0.9	0.65	0.65	0.65	0.5	0.65	0.4
Mid-chord distance from Fuselage (m)	0.9	2.7	4.25	5.55	6.85	8	9.15	10.2
End-chord length (m)	1.31	1.62	1.85	2.08	2.30	2.47	2.70	2.70
Mid-chord length (m)	1.16	1.47	1.74	1.96	2.19	2.39	2.59	2.70
Area of the strip (m^2)	2.08	2.64	2.26	2.55	2.84	2.39	3.36	2.16
Thickness (m)	0.14	0.18	0.21	0.24	0.26	0.29	0.31	0.32
Fuel tank length (m)	0.46	0.59	0.69	0.79	0.88	0.96	1.03	1.08
Fuel tank volume (m^3)	0.00	0.00	0.19	0.24	0.30	0.00	0.42	0.28
Fuel tank volume (m^3) - Integration	**0.000**	**0.000**	**0.189**	**0.241**	**0.299**	**0.000**	**0.418**	**0.280**
Fuel volume inside tank (m^3)	0.000	0.000	0.189	0.241	0.299	0.000	0.418	0.280
Fuel mass inside tank (kg)	0.00	0.00	169.73	216.59	269.16	0.00	376.16	251.94
Fuel weight inside tank (kN)	**0.00**	**0.00**	**4.51**	**5.76**	**7.16**	**0.00**	**10.00**	**6.70**
Empty wing weight (kN)	**2.72**	**3.45**	**2.95**	**3.33**	**3.72**	**3.12**	**4.39**	**2.82**
Extra weights (kN)	0.00	0.00	0.00	0.00	0.00	12.20	0.00	0.00
Total wing weight (kN)	2.72	3.45	7.46	9.09	10.87	15.31	14.39	9.52
Lift (kN)	**18.12**	**23.01**	**19.66**	**22.21**	**24.77**	**20.79**	**29.28**	**18.81**
Resultant force (kN)	**15.40**	**19.56**	**12.20**	**13.12**	**13.90**	**5.48**	**14.89**	**9.29**
Shear force (kN)	**15.40**	**34.96**	**47.16**	**60.29**	**74.18**	**79.66**	**94.55**	**0.00**
Bending Moment (kNm)	**13.86**	**59.18**	**112.56**	**182.40**	**269.81**	**346.73**	**459.96**	**0.00**
Reference axis location	0.46	0.59	0.69	0.79	0.88	0.96	1.03	1.08
Aerodynamic centre location	0.29	0.37	0.43	0.49	0.55	0.60	0.65	0.68
Pitching moment about ref axis (Nm)	831.36	1341.07	1355.74	1730.55	2151.06	-10224.90	3006.96	0.00
Local Twist (kNm)	0.83	1.34	1.36	1.73	2.15	-10.22	3.01	0.00
Total Twist (kNm)	**0.83**	**2.17**	**3.53**	**5.26**	**7.41**	**-2.82**	**0.19**	**0.00**

Table 9: Analysis of Shear Force, Bending Moment and Twist on the wing for the 2.71g case